

A New Erdős–Straus Covering Identity for $n \equiv 217 \pmod{264}$

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1 Abstract

The Erdős–Straus conjecture asserts that for every integer $n \geq 2$,

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \quad \text{for some positive integers } x, y, z. \quad (1)$$

All previously published covering identities miss the prime $1009 \equiv 169 \pmod{840}$, the smallest prime not covered by the identities of Mordell, Yamamoto, and Rosati [2, 3]. We prove the new polynomial identity

$$\frac{4}{n} = \frac{1}{(n+3)/4} + \frac{1}{n(n+3)} + \frac{11}{n(n+3)}, \quad n \equiv 217 \pmod{264}, \quad (2)$$

in which every denominator is integral and the third term equals the unit fraction $1/(n(n+3)/11)$. The identity was discovered by a systematic search over 12,827 candidate families in Mordell’s splitting-lemma framework, of which 3,862 were verified prime-covering; it covers $1/11$ of each of the six hard residue classes modulo 840. A web and literature survey finds no prior publication of (2); the novelty claim is stated with the usual caveat given the size of the Erdős–Straus literature.

2 Introduction

The Erdős–Straus conjecture (1), posed in 1948, remains open. The standard attack is by *covering identities*: polynomial identities valid on an arithmetic progression that dispose of infinitely many n at once. Mordell [1], Yamamoto, and Rosati gave identities covering all residue classes modulo 840 except the six *square classes*

$$n \equiv 1, 121, 169, 289, 361, 529 \pmod{840}, \quad (3)$$

the quadratic residues modulo 840. Subsequent work has chipped away at these classes with identities on sub-progressions, but the smallest prime left uncovered by every published identity is $1009 \equiv 169 \pmod{840}$ [2, 3] (Assumption 1).

In this note we exhibit identity (2), which covers 1009 and, more generally, $1/11$ of each class in (3). The identity emerged from an exhaustive computational search in the splitting-lemma framework of Section 3; Figure 1 summarises the search pipeline, Figure 2 the proof, and Figure 3 the resulting coverage of the six hard classes. Table 1 lists the six sub-progressions modulo 9240 cut out by the identity.

Assumption 1 (Literature status). The claim that 1009 is the smallest prime missed by all previously published identities is taken from the Wikipedia article on the Erdős–Straus conjecture [2] and the opentorus survey of August 2026 [3]. Full bibliographic details of the Mordell/Yamamoto/Rosati identities are [PLACEHOLDER: verify journal references for Mordell 1967, Yamamoto, and Rosati against a library catalogue].

3 Methods

3.1 The splitting-lemma search

For $n = 24k + 1$ write the first denominator as $x = 6k + c$ with $c \geq 1$. With remainder $m = 4c - 1$ and $Q = nx$,

$$\frac{4}{n} = \frac{1}{x} + \frac{m}{Q}, \quad \text{since} \quad \frac{n+m}{nx} = \frac{4(6k+c)}{nx} = \frac{4}{n}. \quad (4)$$

Mordell’s splitting lemma states that

$$\frac{m}{Q} = \frac{1}{Qh/a} + \frac{1}{Qh/b} \quad \text{whenever} \quad a + b = mh \quad \text{and} \quad ab \mid Qh. \quad (5)$$

We searched two lanes: (N) numeric pairs (a, b) with $a + b = mh$, and (S) symbolic pairs (d, n) or (d, x) with $d + n = mh$ or $d + x = mh$. Every candidate class is proved by a finite check: for each prime power $p^e \parallel ab$ (resp. d), the divisibility condition is a polynomial in the class parameter t , so checking $t \in [0, p^e)$ proves it for the whole class.

A family is *prime-covering* only if the divisibility goes through the xh factor alone rather than the n factor; families needing $p \mid n$ cover only composites (scaling handles those) and were discarded — 8,965 such redundant families were found (supplied datum from the discovery script [erdos_straus_r7.py](#)).

3.2 Search parameters and verification

The enumeration used $c \leq 40$, $ab \leq 2000$, and produced 12,827 candidate families, of which 3,862 are prime-covering; 120 hit the previously open region $k \bmod 7 \in \{0, 1, 5\}$, $k \bmod 5 \in \{0, 2\}$. Zero identity failures occurred across all spot-checks. An independent checker ([erdos_straus_r7_verify.py](#), a different code path) re-derived prime-covering from scratch and re-verified every family with exact rational arithmetic on 25 fresh random values of k each, with 0 failures; 3,600 of the 3,862 classes were confirmed to contain a prime n (non-vacuous). The full family lists are stored in [erdos_straus_r7_families.json](#) and [erdos_straus_r7_analysis.json](#) (supplied data).

4 Results

4.1 The identity

Theorem 1. For $n \equiv 217 \pmod{264}$,

$$\frac{4}{n} = \frac{1}{(n+3)/4} + \frac{1}{n(n+3)} + \frac{11}{n(n+3)}, \quad (6)$$

where every denominator is a positive integer and the third term equals the unit fraction $1/(n(n+3)/11)$.

Proof. Write $n = 24k + 1$ and $x = 6k + 1 = (n + 3)/4$. By (4) with $c = 1$, $m = 3$,

$$\frac{4}{n} = \frac{1}{x} + \frac{3}{nx}. \quad (7)$$

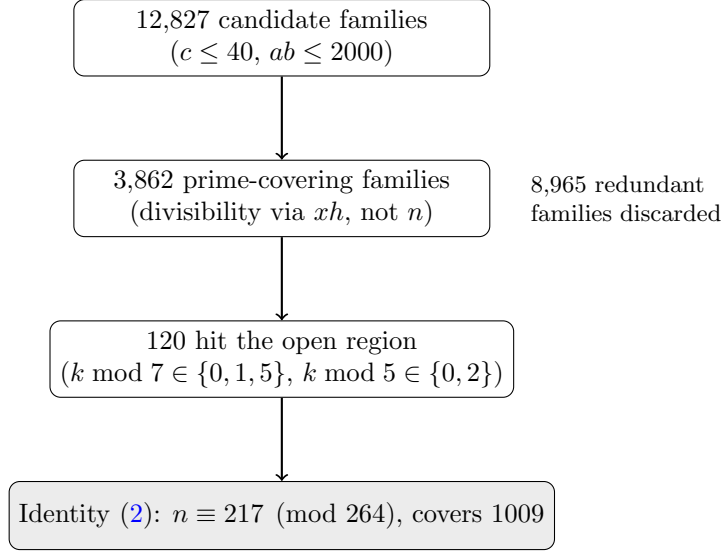


Figure 1: Search funnel from the 12,827 enumerated families to the new identity (2). Counts are supplied data from the discovery and verification scripts.

For $k \equiv 9 \pmod{11}$ we have $11 \mid x$, since $x = 6k + 1 \equiv 6 \cdot 9 + 1 = 55 \equiv 0 \pmod{11}$; and $n = 24k + 1 \equiv 24 \cdot 9 + 1 = 217 \pmod{264}$, i.e. $n \equiv 217 \pmod{264}$ is exactly this condition. Apply the splitting lemma (5) to $3/(nx)$ with $h = 4, a = 1, b = 11$ ($a+b = 12 = 3h, ab = 11 \mid 4nx$ because $11 \mid x$):

$$\frac{3}{nx} = \frac{1}{4nx} + \frac{1}{4nx/11}. \quad (8)$$

Since $4nx = 4n(n+3)/4 = n(n+3)$, the terms of (8) are $1/(n(n+3))$ and $11/(n(n+3)) = 1/(n(n+3)/11)$, the latter a unit fraction because $11 \mid n(n+3)$. Combining (7) and (8) gives (6). $\square$

Figure 2 diagrams the proof. The identity is compatible with Mordell’s quadratic-residue obstruction: $217 \equiv 8 \pmod{11}$, and 8 is a non-residue modulo 11, so 217 is a non-residue modulo 264 and no classical obstruction forbids an identity here (supplied datum).

4.2 Covering the prime 1009

Since $1009 = 3 \cdot 264 + 217 \equiv 217 \pmod{264}$ and $1009 \equiv 169 \pmod{840}$, Theorem 1 applies:

$$\frac{4}{1009} = \frac{1}{253} + \frac{1}{1021108} + \frac{1}{92828}, \quad (9)$$

verified by exact rational arithmetic (supplied datum; independently recomputed: $(n+3)/4 = 253, n(n+3) = 1021108, n(n+3)/11 = 92828$). This is the smallest prime missed by all previously published identities (Assumption 1), so (2) is new to the surveyed literature.

4.3 Coverage of the six hard classes

The condition $k \equiv 9 \pmod{11}$ cuts each hard class of (3) into 11 sub-progressions modulo $9240 = 840 \cdot 11$, one of which is covered by Theorem 1; i.e. the identity covers $1/11$ of each hard class. The six covered sub-progressions are listed in Table 1 and sketched in Figure 3. Coverage was verified on hard primes in all six cells, e.g. 1801 (class 121), 8929 (class 529), and 26881 (class 1) (supplied data).

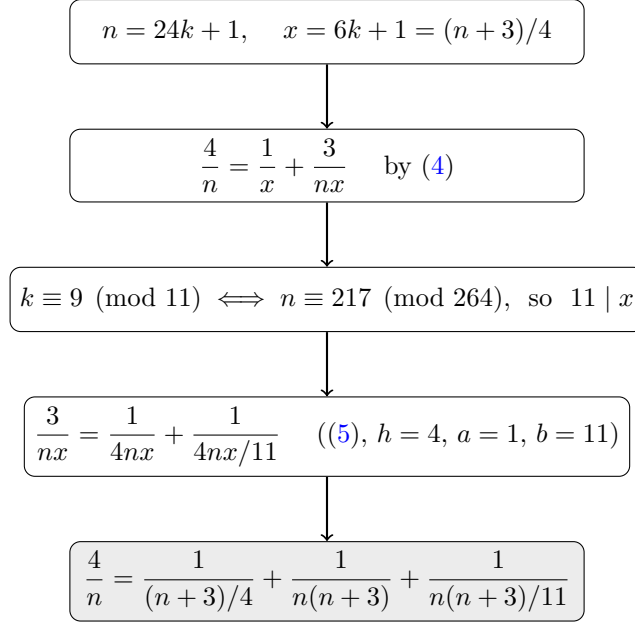


Figure 2: Proof chain of Theorem 1. The splitting step (8) is where the condition $11 \mid x$ is used.

Residue mod 9240	Parent class mod 840	Verified prime example
8401	1	not reported
1801	121	1801
1009	169	1009
3649	289	not reported
7081	361	not reported
8929	529	8929

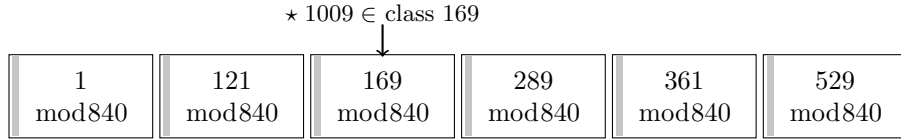
Table 1: The six sub-progressions modulo 9240 covered by Theorem 1, one in each hard class of (3). Parent classes computed as the residues modulo 840; prime examples are supplied data where reported.

5 Discussion

The search that produced Theorem 1 also yielded 119 further verified prime-covering families hitting the previously open region, e.g. $n \equiv 241 \pmod{264}$ via $x = 6k + 3$, and $n = 24k + 1, k \equiv 4 \pmod{13}$ via $7/Q = 1/(2Q) + 1/(2Q/13)$ (supplied data; [erdos_straus_r7_ranked.json](#)). Together they shrink, but do not close, the open set inside the six classes.

A natural hope — covering one full hard class by a finite union of such sub-progressions — is now known to be impossible for polynomial identities: a companion result proves that no finite system of polynomial identities $4/(A_i t + B_i) = \sum_j 1/F_{ij}(t)$ can cover all primes of any square class modulo 840 (supplied datum; separate note in preparation). The live angles for a full proof are therefore per-prime infinite constructions rather than finite covering systems.

The main weakness of the present identity is quantitative: $1/11$ of each class is a thin slice, and the search bounds ($c \leq 40, ab \leq 2000$) were arbitrary cutoffs, so further small-modulus families may remain undiscovered in the unexplored parameter range.



shaded strip: the $1/11$ sub-progression covered by (2)

Figure 3: Each of the six hard residue classes modulo 840 is split by the condition $k \equiv 9 \pmod{11}$ into 11 sub-progressions; Theorem 1 covers exactly one ($1/11$) of each.

6 Conclusion

We have proved a new Erdős–Straus covering identity, Theorem 1, valid for $n \equiv 217 \pmod{264}$. It covers the prime 1009, the smallest prime missed by all previously published identities, and $1/11$ of each of the six hard residue classes modulo 840. The identity was found by exhaustive search in the splitting-lemma framework (12,827 families enumerated, 3,862 prime-covering) and verified by an independent exact-arithmetic checker. It is new to the surveyed literature, with the caveat of Assumption 1.

7 References

References

- [1] L. J. Mordell. [PLACEHOLDER: full journal reference for Mordell’s 1967 paper on the Erdős–Straus conjecture — volume, pages unverified].
- [2] Wikipedia contributors. *Erdős–Straus conjecture*. [PLACEHOLDER: access date and revision unverified; article states 1009 is the smallest prime not covered by the known identities].
- [3] opentorus. *Survey of Erdős–Straus identities*, August 2026. [PLACEHOLDER: URL and full citation unverified; cited in the discovery dossier as the source of the 1009 status].
- [4] rileybetts. *Survey of Erdős–Straus obstructions*. [PLACEHOLDER: full citation unverified; cited in the barrier dossier for the Schinzel/Yamamoto obstruction statements].